

What is Agile?

“Agile is an Iterative and Incremental software development approach where the project is developed in small iterations called sprints. It focuses on flexibility, collaboration, continuous improvement, continuous delivery, and customer feedback.”

- **Iterative** means repeating the process again and again with improvements.
- **Incremental** means developing and delivering the software feature by feature

Agile Principles

1. Customer Involvement

Customer is involved throughout the project by giving continuous feedback.

2. Iterative and Incremental Development

Software is developed and delivered in small parts called sprints.

3. Team Collaboration

Business people, customers, testers, and developers communicate and work together to achieve project goals.

4. Flexibility

Requirement changes are accepted at any stage of development.

5. Frequent Delivery

Working software is delivered regularly in short releases.

6. Continuous Improvement

Team continuously reviews and improves the process based on feedback.

Advantages of Agile

- Faster delivery of software
- Flexible to changes
- Better customer satisfaction
- Better quality product
- Early detection of issues
- Improved team collaboration
- Continuous improvement

Disadvantage:

Less focus on design and documentation since we deliver software very faster.

Scrum

Scrum is an Agile framework where the work is divided into small parts called sprints, and each sprint delivers a working feature of software.

*It is a framework through which we build software product by following agile principles

Scrum includes group of people called scrum team. Normally it contains 5-9 members.

It contains-1.product owner

2.scrum master

3.Development team

Product Owner

“Product Owner is responsible for managing requirements, maintaining the product backlog, prioritizing tasks, and communicating with customers and the Scrum team.”

Scrum Master

“Scrum Master is responsible for ensuring Agile principles are followed properly, conducting Scrum meetings, removing blockers, and helping the team work effectively.”

Development team

Design,Develop and test the software

Scrum terminologies

User Story

“User Story is a small requirement written from the user’s perspective.”

Example of User Story

“As a customer, I want to add products to cart so that I can buy them later.”

Epic

“Epic is a large user story that can be divided into multiple smaller user stories.”

Example

Epic: Online Shopping

User Stories:

- Add to cart
- Online payment
- Order tracking
- Wishlist feature

Epic is a collection of related user stories.

Product Backlog

“Product Backlog is a prioritized list of all project requirements, features, enhancements, and bug fixes that need to be developed in the product. It is maintained by the Product Owner

Sprint/Iteration

“Sprint is a fixed short time period, usually 2–4 weeks, in Scrum where the team completes a set of tasks and delivers a working software.

Sprint Planning

During Sprint Planning, the team discusses:

- Sprint goals
- User stories to be picked from the product backlog
- Task estimation
- Team capacity and workload

At the end of the meeting, the Sprint Backlog is finalized.

Sprint backlog

Sprint Backlog is the list of tasks selected from the Product Backlog for the current sprint that the Development Team commits to complete.

Daily stand-up meeting

In Daily stand-up meeting team members discuss:

- What they completed yesterday
- What they will do today
- Any blockers they are facing

Story Point

Story Point is an estimation technique used to estimate the effort and complexity of a user story.

Fibonacci series, Planning poker are used

Sprint Review

Sprint Review is a meeting conducted at the end of the sprint to show the completed work to the customer or stakeholders and get feedback.”

Sprint Retrospective

Sprint Retrospective is a meeting conducted after sprint completion to discuss

- what went well
- what went wrong
- and areas for improvement.

Definition of Ready (DoR)

“Definition of Ready means a user story or task is clear, understood, and ready to start development.”

Definition of Done (DoD)

“Definition of Done means a task is completely finished, tested, reviewed, and ready for deployment.”

Burndown Chart

“Burndown Chart is used to track how much work is remaining in the sprint.”

Burnup Chart

“Burnup Chart is used to track how much work is completed in the project.”

Agile Roles

- Product owner
- Scrum master
- Development team

Artifacts

- Product backlog
- Sprint backlog
- Burndown chart

Agile Ceremonies

- Sprint planning
- Daily stand-up
- Sprint Review
- Sprint Retrospective

Scrum Board

“Scrum Board is a visual board used to track the progress of tasks during a sprint.”

Common Columns

- To Do
- In Progress
- Testing
- Done